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6. Normal Distribution

1. What is the Normal Distribution?

→ A symmetric, bell-shaped curve where:

- Mean = Median = Mode (center point).
- 68% of data falls within 1 standard deviation (σ) of the mean.
- 95% within 2σ
- 99.7% within 3σ

Example:

Human heights, measurement error, and IQs scores often follow this pattern.

2. What is Z-score: Mathematical Definition and Python Code.

2.

(A) Mathematically

The Z-score measure how many standard deviations a data point is from the mean.

$$Z = \frac{X - \mu}{\sigma}$$

- X : Individual data point
- μ : Mean of the population.
- σ : Standard deviation of the population.

Interpretation

- $Z = 0$: Exactly at the mean.
- $Z = 1.5$: 1.5 standard deviation above the mean.
- $Z = -2$: 2 standard deviation below mean.

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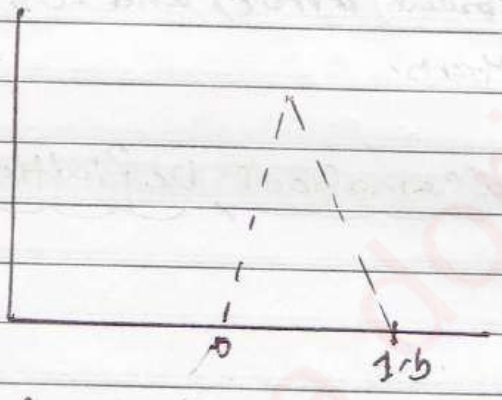
Graphical Interpretation

① Center ($z = 0$)

↳ Location: Exactly at the mean (μ).
↳ Example: Average test score
↳ Visual: Highest point of the curve.
↳ % of data: 0th percentile (exactly average).

② $z = 1.5$ (Above Average)

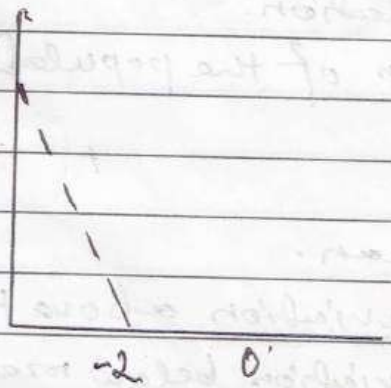
↳ location: Right of center, between $+1\sigma$ and $+2\sigma$
↳ Meaning: Your value is 1.5σ higher than average.
↳ visual:



↳ % of data: ~ 93rd percentile (better than 93% of values).

③ $z = -2$ (Below Average)

↳ location: Half of center, at -2σ
↳ Meaning: Your value is 2σ lower than average.
↳ visual:



↳ % of data: ~ 2nd percentile (only 2% are lower).

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Standard Normal Distribution ($\mu = 0$, $\sigma = 1$)



How to Read the Positions

- Between -1σ and $+1\sigma$: 68% of all data (typical values).
- Beyond $\pm 2\sigma$: Rare values (either exceptional or concerning).
- Beyond $\pm 2\sigma$: Rare values (either exceptional or concerning).
- Beyond $\pm 3\sigma$: Extremely rare ($< 0.3\%$ of cases).

2.

B. Practical Applications:

(1) Outlier Detection:

- Value with $|z| > 3$ are likely outliers.

(2) Comparing Different Scales : SAT vs ACT

Given information

Test	Mean(μ)	Std (σ)	Your Score:
Sat	1000	200	1200
ACT	20	6	30.

Formula: $z = \frac{\text{Your Score} - \mu}{\sigma}$

For SAT (1200):

$$z_{sat} = \frac{1200 - 1000}{200} = \frac{200}{200} = 1.0$$

For ACT (30):

$$z_{ACT} = \frac{30-20}{6} = \frac{10}{6} \approx 1.67$$

Interpretation:

Test	z-score	Percentile (Approx)	Meaning
SAT	1.0	~84%	Better than 84% of test-takers.
ACT	1.67	~95%	Better than 95% of test-takers.

Conclusion

- The ACT score ($z \approx 1.67$) is stronger than the SAT score ($z = 1.0$).
- A z-score of 1.67 is significantly above average.

Key Takeaways

1) z-scores standardize comparisons across different tests/scales.
2) Higher z-score = Better performance relative to peers.
3) ACT ($z = 1.67$) > SAT ($z = 1.0$) in this case.

Rule of Thumb:

- $z > 1.96$ → Top 5%.
- $z > 3$ → Extremely rare ($< 0.1\%$).

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(3) Standardizing feature for ML.

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
Standardized_data = scaler.fit_transform(data.reshape(-1,1))
```

How Standardizing feature and z-score work together?

Step	z-score (Manual)	Feature Standardization (ML)
purpose	Analyze single data points	Preprocess entire features for ML.
Calculation	Manual math	Automated via StandardScaler.
Output	Single z-value	Transformed feature ($\mu=0, \sigma=1$)
use Case	Statistic/EDA	ML pipelines.

→ This topic we'll discuss later or coming chapters.

(4) key Properties

1) mean of z-scores is always 0.
2) std of z-scores is always 1.
3) Preserves the shape of the original distribution.

(5) When to use z-score:

- ✓ Normalization.
- ✓ Comparing variables on different scales.
- ✓ Algorithms requiring standardized data (e.g. PCA, KNN)

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✗ Avoid for

↳ Highly skewed data (use log transformed first.
↳ Data with extreme outliers (use robust scaling instead)

3 Key concepts of Normal distribution

Term	Definition	Example
Standardize	Convert data to mean = 0, σ = 1	Height 176cm → z-score: 1.5
z-score	How many σ a value is from the mean	z = 2 → 2σ above mean.
QQ-Plot	Checks if data is normal	Points on straight line = normal.

Why Normal distribution Matters (Even for Non-Normal Data).

1. Sample statistics Become Normal (central Limit

What happens:

↳ When you take average ($\bar{x}$) of samples (even from non-normal data)

How it works:

(1) Convert values to z-scores.

$$z = \frac{x - \mu}{\sigma}$$

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(2) ~~Now all values are in "standard deviation units"~~

Why it Matters:

- Enables use of:
  - ↳ Confidence intervals (e.g. "Mean height is 170cm ± 5cm").
  - ↳ p-values in hypothesis tests.

(2) Standardization (Apples-to-Apples Comparison)

What it solves:

- Comparing values on different scales (e.g. SAT vs ACT, temperature in °C vs °F).

How it Works:

(1) Convert values to z-scores:

$$z = \frac{x - \mu}{\sigma}$$

(2) Now all values are in "standard deviation units".

Example of ACT & SAT we have already done before.

(3) Model Assumptions (Why Algorithms Care).

Algorithm that need Normalized Data:

(1) ~~Linear Regression~~

- ↳ Assumes residuals (errors) are normally distributed.

(2) ~~Logistic Regression~~.

- ↳ Works better with standardized features.

(3) Neural Networks.

- ↳ Converge faster when inputs are standardized

stoppers

What if You ignore this?

- poor performance
- Misleading feature importance

Key Takeaways

(1) CLT is Magic: Average behave normally → reliable statistics.

(2) Z-scores are Universal: Compare anything (SAT vs ACT, temp, currencies).

5 The Misconception:

✗ Myth: "Most real-world data is normal"

✓ Truth: Raw data (incomes, web traffic) is usually skewed or heavy-tailed.

→ Normality is useful for statistics about samples, not raw data itself.

6. When to Use (and Not Use) Normal Assumptions

✓ Use when:

↳ working with sample means (CLT applies).
↳ Data is approximately symmetric, unimodal.

✗ Avoid when:

↳ Data has extreme outliers or skew
↳ Sample size is tiny ($n < 15$) → use bootstrapping instead.

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(7. Real world Analogy).

Goal Scoring Time : Normal vs. Bimodal Distribution

① Normal Distribution Example.

Scenario : A typical match where goals can be scored at any time.

↳ Most goals cluster around 25-65 minutes (middle of the game).

↳ Fewer goals in first 15 minutes (team settling in)

↳ Fewer goals in last 15 minutes (teams may defends Leads).

Why Normal?

↳ forms a bell curve around the mean (~46th minute).

↳ Symmetrical decrease in early/late goals.

Scenario : A match where:

↳ Most goals come either:

↳ Early (15-25') from high pressing

↳ Late (70-80') when teams tire.

↳ Almost no goals in middle period.

Why Bimodal?

↳ Two distinct peaks at 20' and 76'

↳ valley in between (40-60') with few goals.

Key differences

Feature	Normal distribution	Bimodal Distribution
Peaks	1 (centre)	2 (early + late)
Tactics	Balanced play	Extreme strategy
Analysis	Use mean & std	Require segmentation

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Analytic Insights

- Normal: Predict goals using single normal model.
- Bimodal: Build separate models for early/late phases.

→ This shows how distribution shape affects analysis whether you're studying sports, business metrics, or scientific data.

Key Takeaways:

1. Normal ≠ Common: It's a tool, not a universal truth.
2. Standardization enables comparison across different metrics.
3. Always verify with plots (histograms, QQ-plots) before assuming normality.

Why QQ-Plots Are Essential For Checking Normal Distributions.

1. The Core Purpose of QQ-Plots

→ A Q-Q (Quantile-Quantile) plot is the gold standard for visually assessing if your data follows a normal distribution. It directly compares your data's quantiles to the quantiles of a perfect normal distribution.

→ Key Question it Answers
"How closely does my data match a normal distribution, and where does it deviate?

2) Why QQ-Plots Trump Other Methods.

Methods	Pros	Limitations	QQ-Plot Advantage
Histograms	Simple, intuitive	Bin size affects interpretation	Shows exact deviation points.
Statistical Tests	Provides p-value	Sensitive to sample size	Visual + detailed.
Descriptive Stats	Skew/kurtosis number	Doesn't show shape	Reveals patterns in deviation.

(3) Practical Applications in Data Science

1. → Before Running Parametric Tests.
   - ↳ t-tests, ANOVA, linear regression assume normality.
   - ↳ Example: check residuals in regression models.

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2. Feature Engineering
   - Identify if log / Box-Cox transforms are needed.

3. Anomaly Detection
   - Points far from the line may be outliers.

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Extra Core Conceptions:

1. "Understanding the standard deviation with real world analogy"

→ Imagine you're measuring the travel times of 5 drivers of 5 drivers who each drive a different route in the city. The times they take (in min) are as follows.

- Driver 1: 30 minutes
- Driver 2: 35 minutes
- Driver 3: 40 minutes
- Driver 4: 45 minutes
- Driver 5: 50 minutes.

Step 1: Calculate the Average (Mean).

→ First, you calculate the average travel time. This is the overall "expected" time that reflect how long, on average, it takes the drivers to complete their routes.

$$\text{Average time} = \frac{30+35+40+45+50}{5} = 40 \text{ minutes}$$

So, on average, the drivers take 40 minutes

Step 2: Find the Deviation from the Mean.

Next, Let's look at how each driver's travel time differs from the average. This shows the deviation of each driver's time from the average.

- Driver 1: 30 - 40 = -10 min.
- Driver 2: 35 - 40 = -5 min.
- Driver 3: 40 - 40 = 0 min.
- Driver 4: 45 - 40 = 5 min.
- Driver 5: 50 - 40 = 10 min.

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Step 3: Square the Deviations.

Now, to eliminate negative values (since you can't have negative distance), we square the deviations. This gives us the squared deviations.

- Driver 1: $(-10)^2 = 100$.
- Driver 2: $(-5)^2 = 25$.
- Driver 3: $(0)^2 = 0$.
- Driver 4: $(5)^2 = 25$.
- Drivers: ~~100~~ $(10)^2 = 100$.

So, the squared deviation are: 100, 25, 0, 25, and 100

Step 4: Find the Average of squared Deviations.

To understand the spread, we calculate the average of the squared deviations. This is called the variance

$$\text{Variance} = \frac{100 + 25 + 0 + 25 + 100}{5} = 50.$$

So, the variance of the travel times is 50 minutes^2.

Step 5: Take the square Root of the Variance.

Finally, to bring the units back to minutes (since variance is in minutes^2), we take the square root of the variance. This gives us the standard deviation.

$$\text{std dev} = \sqrt{50} \approx 7.07 \text{ min}.$$

What Does this Mean?

The std dev of 7.07 min tell us how spread out the travel times are around the average time of 40 minutes. In other words, on average, the travel times deviate by about 7.07 minutes from the mean.

- If a driver's time is within 7.07 min of the mean, it's considered typical.
- A time beyond 7.07 min would be considered an

outlier or an exceptional case.

So, in a nutshell,

→ std dev is a way to measure the spread or variability in your data.

→ It is calculated by taking the square root of the variance, which is based on how much each individual data point deviates from the mean.

2) Empirical Rule or 68-95-99.5 rule.

(a) 1 std dev

→ $\mu \pm 1\sigma$

→ 68% of data falls within 1 std dev of the mean.

~~→ 98% of~~

→ Example: 1 std dev from the mean (40 ± 7) would cover travel times from 33 to 47 minutes.

(b) 2 std dev

→ $\mu \pm 2\sigma$

→ 95% of data falls within 2 stand dev of the mean.

→ Example: 2 std devs from the mean (40 ± 2×7). would cover travel time from 26 to 54 minutes.

(c) 3 std dev

→ $\mu \pm 3\sigma$.

→ 99.7% of data falls within 3 std dev of the mean.

→ Example: 3 std dev from the mean (40 ± 3×7) would cover travel time from 19 to 60 minutes.

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8) what are plotted inside the Bell curve

→ Raw data with histogram is plotted.
→ When you create a histogram, you are plotting the raw data - the actual values you collected or simulated
→ The x-axis represents the range of data values. (e.g. travels, heights, exam scores).
→ The y-axis shows how frequently (or how densely) those values occur - either as counts or probability density (0 to 1)

Summary

	what is it?	is it plotted?	where / How?
✓	Raw Data	Yes	As a histogram
✓	Mean (μ)	Not plotted directly, but used	As the center of the curve.
✓	Std dev. (σ)	Not plotted directly, but used	Defines the spread / width of the bell curve.
✗	Variance (σ^2)	No	Only used mathematically to get 'σ'

Empirical Rule

purpose: ~~Unde~~ purpose: understand how data spreads in a normal distributions.

Applies when: Population is normally distributed.

Formula : $\mu \pm (n)\sigma$ $n = 1, 2, 3$.

python code

ways to find z-score.

(1) Manual

```
z_scores = [(x - mean) / std for x in data]
```

(2) Scipy import stats.

```
from
z_scores = stats.zscore(data)
```

(3) printing both

```
print("z-scores:", [round(z,2) for z in z_scores])
```

Act vs Sat

let's code

```
# import numpy as np.

# sample data
data = np.array([12, 15, 18, 22, 24, 25, 30])

# Calculate mean and std
mean = np.mean(data)
std = np.std(data, ddof=1)

# Calculate z-scores.
(1) z_scores = [(x - mean) / std for x in data]

print("Data:", data)
print("Mean:", mean)
```

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print ("std Dev:", std).
print ("z-scores:", [round(z,2) for z in z_scores])

(2)

output

Data: [12, 15, 18, 22, 24, 25, 30]
Mean: 20.857142
Std Dev: 6.225
z-scores: [-1.42, -0.94, -0.46, 0.18, 0.5, 0.69, 1.47]

2-Score Transformed Data.

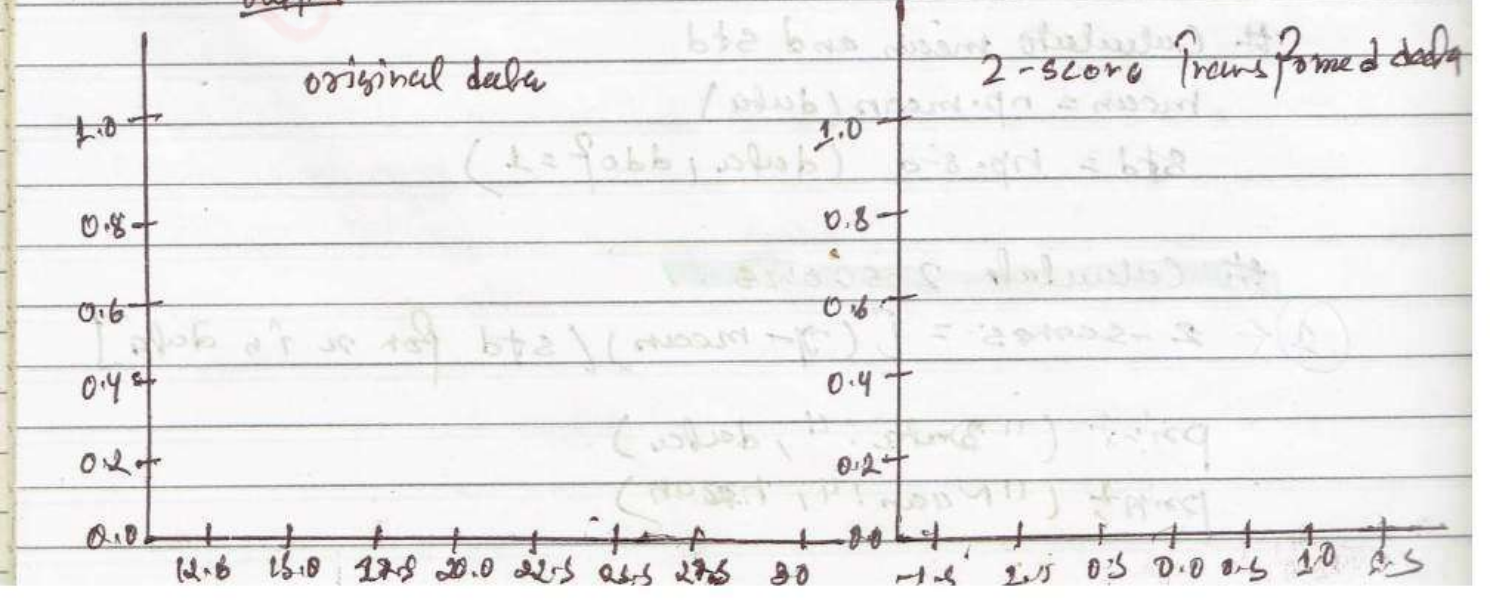
import matplotlib.pyplot as plt

plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
plt.hist(data, bins=7, alpha=0.7).
plt.title("Original Data")

plt.subplot(1,2,2).
plt.hist(z_scores, bins=7, alpha=0.7).
plt.title("z-score Transformed Data").

plt.show().

output



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Act vs sat

```
Import scipy.stats as stats

#SAT
sat_z = (1200 - 1000)/200
sat_percentile = stats.norm.cdf(sat_z) * 100
                 # Convert z to percentiles.

#ACT
act_z = (30 - 20)/6
act_percentile = stats.norm.cdf(act_z) * 100.

print(f"SAT: z = {sat_z:.2f}, Percentile =
      {sat_percentile:.1f}%")

print(f"ACT: z = {act_z:.2f}, Percentile
      = {act_percentile:.1f}%")
```

Output

SAT: z = 1.00, Percentile = 84.1%.
ACT: z = 1.67, Percentile = 95.2%.

1) This line converts a z-score into a percentile rank using cumulative distribution function (CDF) of the normal distribution which we will discuss later

---

→ The original data shows values in their natural units

⇒ → The z-score transformed data shows how many standard deviations each value is from the mean.

→ Typically, z-score transformation centers the data around 0 with most values between -3 and 3.

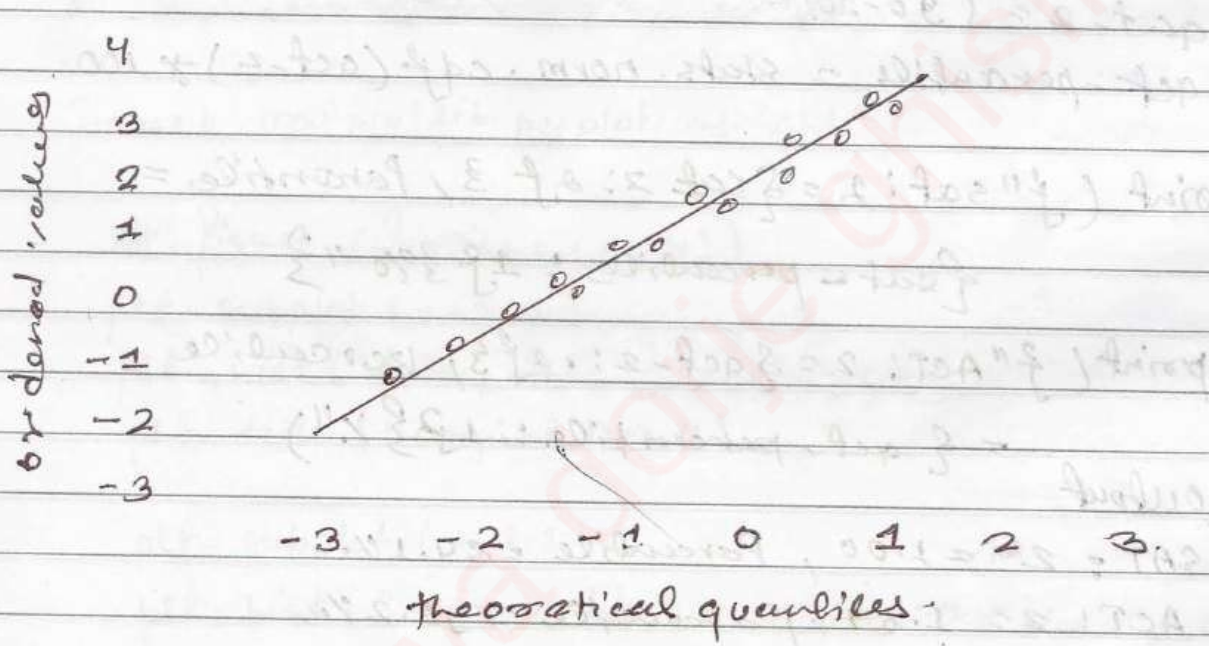
How to Read a QQ-Plot

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(a) Perfectly Normal data

```
Import numpy as np.
Import scipy.stats as stats.
Import matplotlib.pyplot as plt.

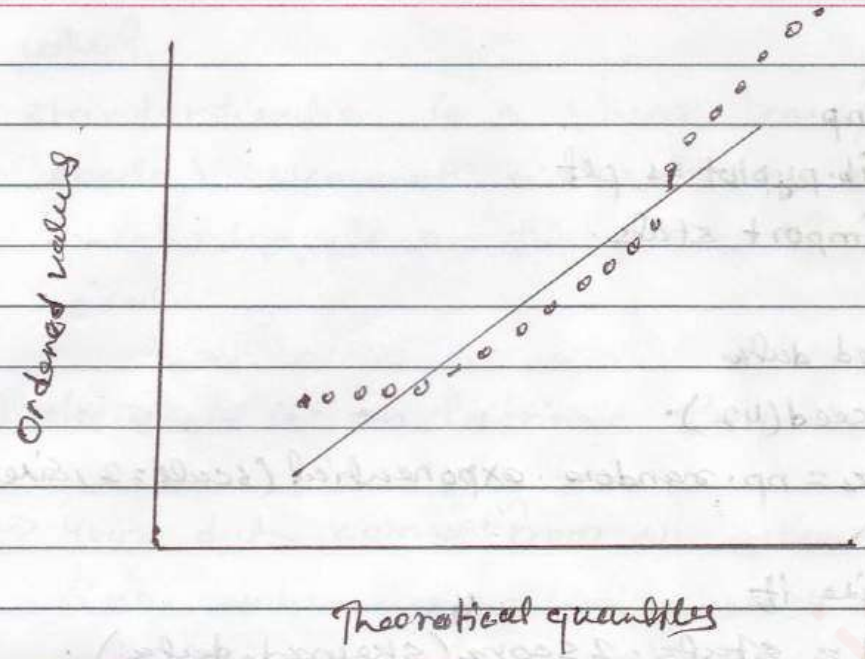
normal_data = np.random.normal(loc=0, scale=1, size=1000)
stats.probplot(normal_data, dist="norm", plot=plt)
plt.title("QQ-Plot: Normal Data")
plt.show()
```



(b) Right-skewed Data (common in Real word).

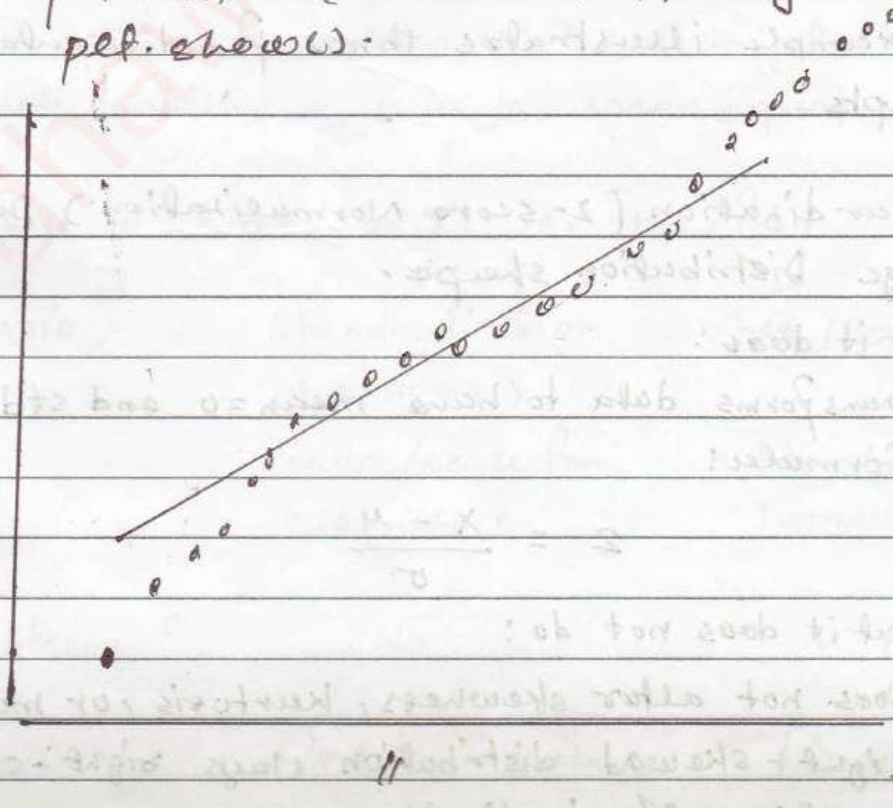
```
Skewed_data = np.random.exponential(scale=1,
stats.probplot(skewed_data, dist="norm", size=1000)
, plot=plt).

plt.title("QQ-Plot: Right-Skewed Data")
plt.show().
```



(c) Heavy-Tailed Data (Outliers Present)

```
heavy_tailed = np.concatenate([np.random.normal
                (size=900), np.random.normal
                (loc=0, scale=5, size=100)])
stats.probplot(heavy_tailed, dist='norm',
               plot=plt)
plt.title("QQ-Plot: Heavy-Tailed Data")
plt.show()
```



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Standardization is not equal to normalization

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

# Create skewed data
np.random.seed(42)
skewed_data = np.random.exponential(scale=2, size=1000)

# Standardize it
z_skewed = stats.zscore(skewed_data)

# Plot comparison
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12,5))
ax1.hist(skewed_data, bins=30)
ax2.set_title("Original skewed Data")
ax2.hist(z_skewed, bins=30)
ax2.set_title("Standardized Data (still skewed)")
plt.show()
```

→ This example illustrates three fundamental statistical concepts.

(1) Standardization (z-score Normalization) Does Not Change Distribution shape.

a) what it does:

→ Transforms data to have mean = 0 and std dev = 1

→ Formula:

$$z = \frac{x - \mu}{\sigma}$$

b) What it does not do:

→ Does not alter skewness, kurtosis, or modality

→ Right-skewed distribution stays right-skewed after standardization.

why?

Standardisation is a linear transformation (shift + scale). skewness is a measure of asymmetry, which depends on the orders of values, not their scale.

(2) skewness is an intrinsic Property of the Distribution.

→ Your data comes from an exponential distribution (np. random.exponential), which is:
- ↳ Always right-skewed (long tail to the right).
- ↳ Has a sharp peak near zero.

→ Standardising shifts and scales the data but preserves the skewness.
- ↳ Before: Mean = 2, SD = 2 (exponential with scale = 2)
- ↳ After: Mean = 0, SD = 1, but skewness = 2 (same as original)

Visual Check:
Both histograms show the same right-tailed shape, just centered at 0 in the second plot.

(3) When to Standardise vs. Transform.

Scenario	Standardisation (z-score)	Log / Box-cox transform
Goal	Centers/scale for algorithms	Fix skewness for normality
Change skewness?	✗ No	✓ Yes.
Use Cases	PCA, SVM, K-Means	Linear regression, t-tests.

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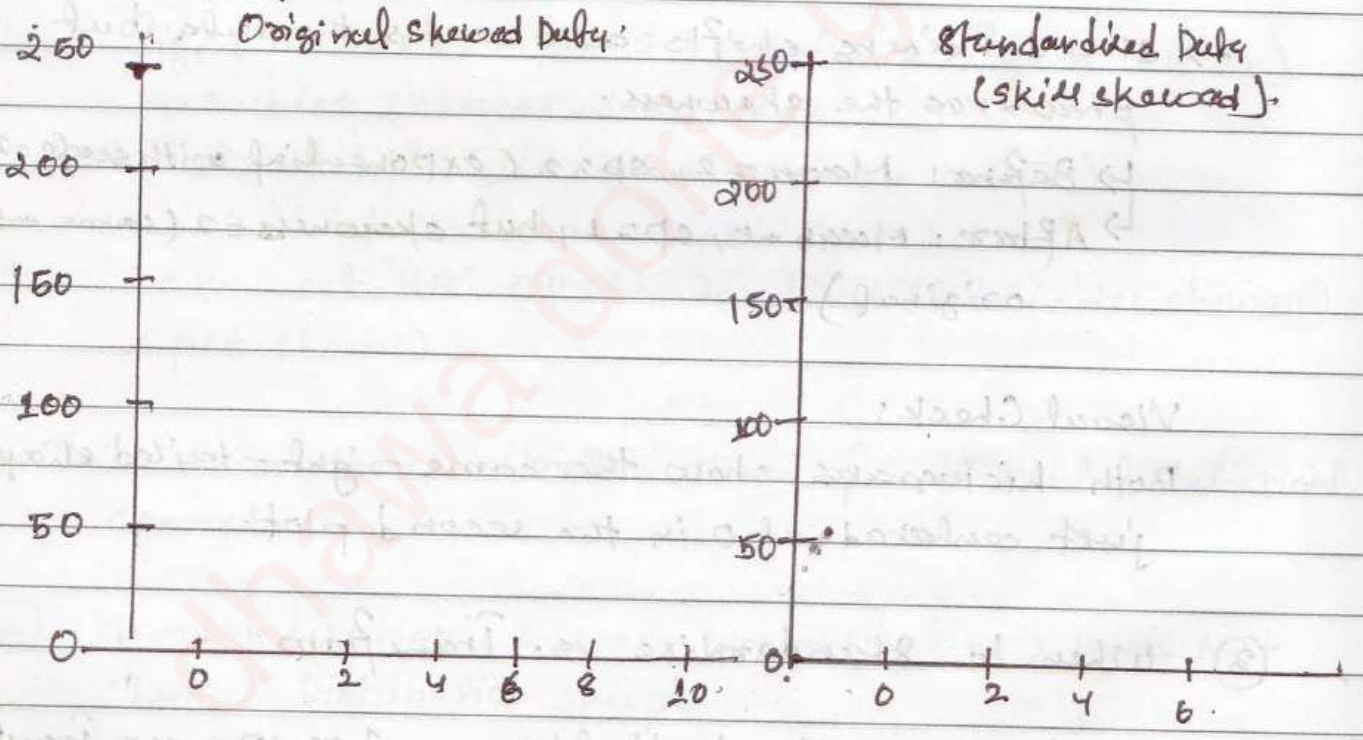
Key Takeaways

(1) Standardization ≠ Normalization.
↳ Use standardization for scaling, not fixes skewness

(2) Skewness requires nonlinear transforms.
↳ log, square root, or Box-Cox for right-skewed data

(3) Always visualize
↳ Histograms and Q-Q plots reveal skewness even after standardization.

output



Why Standardisation Doesn't change distribution shape?

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(1)
⇒ because it's linear transformation.

→ Formula

$$z = \frac{x - \mu}{\sigma}$$

↳ subtracting μ (mean): shifts the distributions left/right (centers it at 0)

↳ Dividing by σ (std dev): stretches/squeezes the spreads (sets $\sigma = 1$).

→ These operation don't alter
- → skewness
- → kurtosis.
- → Modality (no. of peaks)
- → Relative distance between data points.

(2) Why shapes persists

→ Order Preservation: if $x_1 < x_2$ originally: then $z_1 < z_2$ after standardisation.

→ Relative Distances: The gap between points scales proportionally (e.g. outliers remain outliers).

. Examples:

Original Data (x)	Standardised (z)
10	(10 - 20)/5 = -2
20 (mean)	0
30	+2
100 (outlier)	+16

The outlier at 100 is still extreme after standardisation.

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Key takeaways

(1) Standardization is linear:
→ only shifts and scales data.

(2) Shape preservation matters:
↳ Helps you compare variables on a common scale.
↳ Maintains underlying data structure.

(3) To alter shape
→ use non linear transformation (log, Box Cox)

Analogy

→ Standardizing is like converting F to C
↳ Temperature values change, but weather patterns (distribution shape don't).

→ Transforming is like switching from linear to logarithmic scale.
↳ Both values and pattern change.

⑤ Standard Normal and QQ-Plots.

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1. Standard Normal Distribution (Z-Distribution)

- Definition: Normal distribution with $\mu = 0$ and $\sigma = 1$
- How to standardize:

$$z = \frac{x - \mu}{\sigma}$$

- Key Properties
  - ↳ 68% of data within $\pm 1\sigma$
  - ↳ 95% of within $\pm 2\sigma$
  - ↳ 99.7% of within $\pm 3\sigma$

Python Standardization

```
from scipy import stats.
z_scores = stats.zscore(data)
```

2. QQ-Plots: Visual Normality Check.

- X-axis: Theoretical normal quantiles.
- Y-axis: Your data's quantiles.
- Perfect normality: Points fall on the 45° line

How to Create:

python

```
import matplotlib.pyplot as plt
stats.probplot(data, dist="norm", plot=plt)
plt.show().
```

Interpretation Guide:

pattern	Indicates	Action
Straight line	Normal	Use parametric tests.
S-shaped curve	Heavy tails	Try t-distribution.

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C-Skewed curve — Skewness — Apply log transformation

Outliers — Extreme values — Investigate data quality

(3. Key Warnings About Standardization and QQ-Plots.)

(1.) Standardization ≠ Normalization.

What happens

↳ Standardization centers and scales your data ($\mu = 0, \sigma = 1$).

↳ It does change the fundamental shape of your distribution. The right tail remains stretched out. Standardization only changed the axis numbers, (axis data = std dev (z-score scale)), not

(2.) QQ-Plot Limitations and Best Practices.

(a) Small sample warning ($n < 50$).

↳ Even normal data may look non-normal in QQ-plots with small n.

(b) Always Pair with Histograms

↳ Three-Panel Diagnostic Approach.

↳ (A) Raw Histogram: Initial shape assessment.
↳ (B) Standardized Histogram: Verify scale independence
↳ (C) QQ-plot: Detailed deviation analysis.

QQ-Plot limitations and Best Practices

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(a) Small Sample warning (n < 50)

→ Even genuinely normal data can look "non-normal" in a QQ-Plot when the sample size is small.

→ Small samples are noisy.

```
Import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

# small genuine normal sample
small_sample = np.random.normal(size=30)

plt.figure(figsize=(12,5))

# Histogram
plt.subplot(121)
plt.hist(small_samples, bins=10, colors=
        'skyblue', edgecolor='black')
plt.title("Histogram (n=30)")

# QQ plot
plt.subplot(122)
stats.probplot(small_samples, plot=plt)
plt.title("QQ-Plot (n=30) - can be Misleading")

plt.tight_layout()
plt.show()
```

Key points

⇒ Random fluctuations can cause QQ-plot points to drift off the line

⇒ Don't panic unless pattern deviations are

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(b) Always Pair with Histograms.

→ Histograms + QQ-plots together a more reliable analysis

→ Three-Panel Diagnostic Approach.

Panel	Purpose.
Raw Histogram	Understand the original shape (e.g. skewed, multi-mode).
Standardized Histogram	check if skewness remains after rescaling.
QQ-plot	Detect tail behavior and detailed deviations.

Types of Z-Test:

(1) One-sample Z-test.
(2) Two-sample Z-test.
(3) Z-Test for Proportion.

(1) One-sample Z-test.

(a) Theoretical Explanation.

Goal: To test whether the mean of a sample significantly differs from a known population mean.

Hypothesis

- Null Hypothesis (H_0): The sample mean equals the population mean (μ_0)
- Alternative Hypothesis (H_1): The sample mean is different from the population mean.

Z-Test Formula:

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Where:

→ $\bar{x}$ = Sample mean
→ μ_0 = Population mean (known or assumed)
→ σ = Population standard deviation (known)
→ n = Sample size

(b) Practical Example:

Problem: Suppose a machine is designed to fill 500 ml of liquid into bottles. We suspect it's underfilling. We collect a sample of 36 bottles, and the sample mean

is 495 ml. The population standard deviation is 10 ml.

Soln

Population mean (μ_0) = 500 ml

Sample mean ($\bar{x}$) = 495 ml

Population std (σ) = 10 ml.

Sample size (n) = 36

Significance level (α) = 0.05 (default, for 95% confidence).

Step 1 State the hypotheses.

- H_0 : $\mu = 500$ (The machine is filling correctly)
- H_1 : $\mu < 500$ (The machine is underfilling)

The is a left-tailed test.

Step 2 Calculate the z-statistics.

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

$$z = \frac{495 - 500}{10/\sqrt{36}} = \frac{-5}{10/6} = \frac{-5}{1.6667} \approx -3.00$$

Step 3: Find the critical z-value

For a left-tailed test at $\alpha = 0.05$, the critical z-values from the z-table is:

$$z_{critical} = -1.645.$$

Step 4: Make a Decision

→ Our Z-statistic = -3.00

→ Critical Z = -1.645.

→ Since -3.00 < -1.645, we reject the null hypothesis.

Conclusion

There is strong statistical evidence that the machine is under filling bottles.

(c) **Python Code**

```
import numpy as np.
from statsmodels.stats.weightstats import ztest
import matplotlib.pyplot as plt.
from scipy.stats import norm.

# known values.
mu_0 = 500
sigma = 10
n = 36.

sample_data = np.random.normal (loc = 495
        scale = sigma, size = n)

# Perform one-sample Z-test.
Z_stat, p_value = ztest (sample_data,
                value = mu_0, alternative
                = 'two-sided')
print (f"Z-statistics: {Z_stat: .3f}").
print (f"p-value: {p_value: .3f}")
```

output

z-statistic : -2.500.

p-value : 0.012.

(d) Visualization

```
x = np.linspace(-4, 4, 1000).
y = norm.pdf(x, 0, 1).
plt.plot(x, y, label = 'Standard Normal Distribution')

# Shade rejection region of 95% confidence
z_critical = norm.ppf(0.975) # Two-tailed,
                             alpha = 0.05.
plt.fill_between(x, y, where = (x <= -z_
    critical) | (x >= z_critical), color =
    'red', alpha = 0.3, label = 'Rejection Region')

# Plot the computed z-stat.
plt.axvline(z_stat, color = 'black', linestyle =
    '--', label = f'z-stat = {z_stat: .2f}')

# Add labels
plt.title("one-sample z-Test")
plt.xlabel("z value")
plt.ylabel("probability Density")
plt.legend().
plt.grid(True).
plt.show().
```

Interpretation

→ If the p-value < 0.05, reject null hypothesis
  → significant difference.
→ The Z-score tells how many standard deviations the sample mean is from the population mean.
→ The visualization shows if the Z-score lies in the rejection region (tails of the curve).

Queries

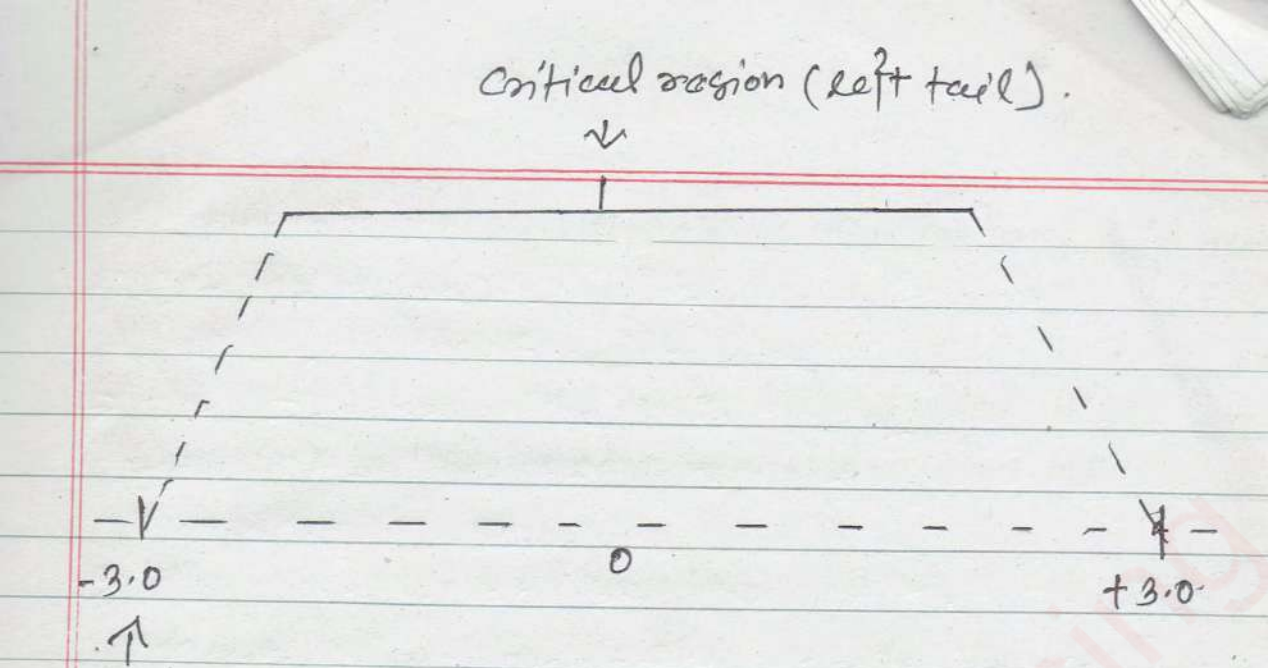
(b) Problems

This is left-tailed test?

→ "left-tailed test" simply means:
  We are testing if the sample mean is significantly less than the population mean.

Why "left-tailed"?

→ In a normal distribution, the "left tail" refers to the lower end of the curve (i.e. values less than the mean)
In your case:
→ Population mean (μ_0) = 500
→ You're checking if the machine is underfilling
  i.e. $\mu < 500$
→ So we are interested in the left tail of the normal curve.



z-statistic falls here → we reject H_0.

Summary:

→ left-tailed test: $H_1 : \mu < \mu_0$ → focus on the left side of the distribution.

→ Right-tailed test: $H_1 : \mu > \mu_0$ → focus on the right side

→ Two-tailed test: $H_1 : \mu \neq \mu_0$ → check both sides for extreme values.

(c) Python queries

2-test using statsmodels

```
z_stat, p_val = ztest(sample_data, value = mu_0, alternative = 'two-sided')
```

↳ **value = mu_0**:

This is the null hypothesis mean (μ_0) i.e. the value you are comparing the sample mean to. In this case, 500 ml

↳ **alternative = 'two-sided'**:

specifies the type of hypothesis test:

- 'two-sided': $H_1 : \mu \neq \mu_0$ → test for any difference (more or less).
- 'smaller': $H_1 : \mu < \mu_0$ → left-tailed test.
- 'larger': $H_1 : \mu > \mu_0$ → right-tailed test.

What z test( ) Returns:

→ **z-stat:** The Z-score, a standardization value showing how far the sample mean is from the population mean, in units of standard error.

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

→ **p-value:** The probability of observing a test statistic as extreme (or more extreme) under the null hypothesis.
if this is < 0.05 (common threshold), you reject H_0.

Output

z-statistic: -2.503

p-value: 0.012.

This suggest the sample mean is significantly lower than 500 - Strong evidence against H_0.

④ Visualization (code breakdown).

① → x = np.linspace(-4, 4, 1000).

↳ Creates 1000 evenly spaced values from -4 to 4 (range of 2 z-scores).

↳ This will be the x-axis for the normal distribution.

② → y = norm.pdf(x, 0, 1).

↳ computes the probability Density function (PDF) of the standard normal distribution N(0,1) for each value in x.

③ → plt.plot(x, y, label = 'Standard Normal Distribution')

↳ Plots the normal bell curve.

↳ Adds a label for the legend.

④ → Rejection Region (Significance level $\alpha = 0.05$).

2-Critical = norm.ppf(0.975).

→ You are performing a two-tailed z-test at significance level $\alpha = 0.05$.

This means:

→ You're testing both:

↳ if the sample mean is too low (left tail)

↳ OR too high (right tail).

→ You want to know whether the sample mean is significantly different from the population mean.

What does "two-tailed" really mean?

→ You're checking for extremes on both ends of the normal distribution.

Area under standard normal curve = 100% or 1.0

$\alpha = 0.05$ → means you're allowing a 5% chance of error.

In a two-tailed test:
- Left tail gets $\alpha/2 = 0.025$ (2.5%)
- Right tail gets $\alpha/2 = 0.025$ (2.5%).

This leaves:
- Middle 95% of the curve = where we don't reject the null
- Outer 5% = the rejection region (split into two)

What does norm.ppf(0.975) do?

→ norm.ppf(p) gives the z-score that corresponds to cumulative probability p

So:
- norm.ppf(0.975) → means:
  "What z-score corresponds to 97.5% of the area to the left?"

And the answer is:
$z \approx 1.96$

Likewise:
- norm.ppf(0.025) → gives $z \approx -1.96$.

So in a two-tailed test with $\alpha = 0.05$:

- You reject it to if your Z is:
  - Less than -1.96.
  - Or greater than $+1.96$.

Why 0.975 and not 0.95?

If you're interested in two-tailed 5% total, then:

So in a two-tailed test with $\alpha = 0.05$

- You reject H_0 if your z is:
  - less than -1.96.
  - or greater than $+1.96$.

Why 0.975 and not 0.95?

if you're interested in two-tailed 5% total then:

- Each tail gets $2.5\% = 0.025$.
- The remaining middle $= 95\% = 0.95$.

But:

- The right tail cutoff is at the 97.5% point (0.025 left tail + 0.95 middle).
- So the upper z-cutoff is at norm.ppf $(0.975) \rightarrow +1.96$.
- The lower z-cutoff is at norm.ppf $(0.025) \rightarrow -1.96$.

Finally Summary

Concept	Value
Significance	0.05.
Tail area	0.025.
norm.ppf (0.025)	-1.96.
norm.ppf (0.975)	$+1.96$
Fail to reject H_0 if	$-1.96 \leq z \leq +1.96$
Reject H_0 if	$z < -1.96$ or $z > +1.96$.

(5) plt.fill_between(x, y, where=(x <= -2.critical) | (x >= 2.critical), color='red', alpha=0.3, Label='Rejection Region').

- Fill both tails beyond ±1.96 in red (α=0.05 rejection regions).
- alpha=0.3 makes the shading semi-transparent.

dhawa doje ghising

2) Two-sample z-test

Objective: To test whether the difference between two sample means is significant or not.

Formula:

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Where:

- $\bar{x}_1, \bar{x}_2$: sample means.
- σ_1, σ_2 : known population standard deviations
- n_1, n_2 : sample sizes.

Hypothesis

- $H_0 : \mu_1 = \mu_2$ (no difference).
- $H_2 : \mu_1 \neq \mu_2$ (there is a difference -- two-tailed)

Practical Example:

Suppose you want to compare two filling machines.

- Machine A: Mean = 498ml, std dev = 10ml, n=36
- Machine B: Mean = 502ml, std dev = 12ml, n=30

Soln

We are testing whether the two machines are significantly different in their mean fill volumes.

- Machine A:
  - Mean $\bar{X}_1 = 498$
  - Std Dev $\sigma_1 = 40$
  - Sample size $n_1 = 36$.

- Machine B:
  - Mean $\bar{X}_2 = 502$
  - Std dev $\sigma_2 = 12$
  - Sample size $n_2 = 30$.

$$z = \frac{498 - 502}{\sqrt{\frac{16^2}{36} + \frac{12^2}{30}}}$$

$$z \approx -1.453$$

Find the P-value

Since this is a two-tailed test, we double the area in one tail:

$$p = 2 \times P(Z > |-1.453|) = 2 \times P(Z > 1.453).$$

Using the standard normal table or scipy.stats.norm.sf()

$$P(Z > 1.453) \approx 0.073 \Rightarrow p \approx 2 \times 0.073$$

$$= 0.146.$$

Final Answer:

- Z-score ≈ -1.453
- p-value ≈ 0.146

Interpretation

At $\alpha = 0.05$:

↳ since $p > 0.05$, we fail to reject the null hypothesis.

↳ There is not enough evidence to say the machines fill differently.

Python code (Manual calculation/there is no built in function).

```
import numpy as np
from scipy.stats import norm.

# sample data
x1_bar = 498
x2_bar = 502
sigma1 = 10
sigma2 = 12
n1 = 36
n2 = 30

# Calculate z-score
numerator = x1_bar - x2_bar
denominator = np.sqrt((sigma1**2/n1) + (sigma2**2/n2))
z_score = numerator/denominator.
```

Calculate two tailed p-value

p-value = 2 * norm.sf(abs(z-score))

print(f"z-score: {z_score:.3f}")
print(f"p-value: {p_value:.3f}").

Visualization of Rejection Region

import matplotlib.pyplot as plt

Standard normal curve.

x = np.linspace(-4, 4, 1000).

y = norm.pdf(x).

Critical values for α = 0.05.

z_crit = norm.ppf(0.975).

plt.plot(x, y, label='Standard Normal')
plt.fill_between(x, y, where=(x <= -z_crit) | (x >= z_crit), color='red', alpha=0.3, label='Rejection Region').

Plot z-score.

plt.axvline(z_score, color='black', linestyle='--', label=f'z = {z_score:.2f}').

plt.title("Two-sample z-test")
plt.xlabel("z value")

```
plt.ylabel("Probability Density")
plt.grid(True)
plt.show()
```

Interpretations

- If p-value $< \alpha$, reject the null → significant difference between machines
- If p-value $> \alpha$, fail to reject the null → no evidence of a difference.

Feature	One-Sample Z-Test	Two-Sample Z-Test
Purpose	Compare sample mean to population mean.	Compare means of two independent samples.
Hypotheses	$H_0 : \mu = \mu_0$	$H_0 : \mu_1 = \mu_2$
Population Std dev	Known	Both groups' Std devs are known
Formula	$z = \frac{\bar{x} - \mu_0}{\sigma \sqrt{n}}$	$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$
Application Example	Is a machine underfilling bottles?	Do two machines fill bottles with different amounts.

one-sample

p-value = 2 * norm.cdf(2-score).

VS

two-sample

p-value = 2 * norm.sf(abs(2-score)).

---

what is a p-value?

A p-value is the probability of getting a test statistic (like your z-score) at least as extreme as the one you observed, assuming the null hypothesis is true.

Real-world analogy:

Imagine you're planning a big outdoor event-like a wedding in the park.

↳ The weather forecast says there's only a 5% chance of rain.

↳ You trust the forecast (this is your null hypothesis: "It won't rain")

↳ Now on the day of the event... it rains.

what just happened?

You think:

| "wait -- the forecast said only a 5% chance. That's rare something fool off.

This situation is like p-value of 0.05.

It means:

| "If the forecast were correct (null is true), the chance of this rain happening is only 5%."

so:

- If it rains, even with a low probability, you might question the accuracy of the forecast (null hypothesis).
- But if you had 20% chance of rain ($p = 0.2$) and it rains, you'd say

> "well, fair enough - the forecast said it's pretty likely."

Low p-value ($\alpha = 0.05$): "Rain despite low odds".

Elements	Meaning
Forecast says only 5% chances of rain (p-value $= 0.05$)	if H_0 is true (sunny day), rain is very unlikely.
You step outside -- and it's pouring.	The actual outcome (data) is very different from what was expected under H_0.
You now doubt the forecast (H_0).	since this outcome is rare under H_0, you consider rejecting it.
Interpretation	This rain was too rare to ignore. Maybe the forecast was wrong. I reject H_0.

Conclusion: You reject the null hypothesis
You believe something else is going on --
like the forecast (machine) being inaccurate

High p-value ($> \alpha = 0.05$): "Rain was likely anyway".

Element	Meaning
Forecast says 40% chance of rain (p-value = 0.40)	Rain is fairly common even if Ho says "sunny"
It rains	Yes, but that's not surprising given the forecast.
You don't doubt the forecast (Ho)	Since rain is quite probable, the data doesn't contradict Ho
Interpretation	"Rain isn't rare in this forecast. Nothing unusual happened."

Conclusion: You fail to reject Ho. The observed data is not surprising - you stay with the forecast.

• cdf(z-score) vs. sf(abs(z-score)).

* Analogy: Bridge analogy to understand p-value and tails in z-test.

* Imagine this: You're standing in the middle of a bridge. That middle point is mean of the standard normal distribution, which is:

| z = 0.

* The Bridge
  - The left end of the bridge is z = -∞
  - The right end of is z = +∞
  - You're standing at the center: z = 0.

* Now imagine you're asking
"How unusual is it to walk just from the center, say z = ±1.5?
That's the idea of a z-statistics.

One-Tailed Test Analogy

* Left-Tailed Test:
  - You walk to the left of the bridge (-ve z)
  - You calculate how much of the bridge lies to the left where you stopped.

| use: p = norm.cdf(z) if z is negative.

* Right-Tailed Test:
  - You walk right, stopping at some z = +1.5

- You look at how much of the bridge lies to the right of you

| use: p = norm.sf(z).
| Equivalent to: 1 - norm.cdf(z).

Two-Tailed Test Analogy

In a two-tailed test, you're saying:

| "I don't care which direction - if you walk too far in either direction, it's suspicious!".

So:

→ You check how much of the bridge lies beyond ±z

→ You multiply that tail area by 2 (because you're checking both sides)

☑ use: p = 2 * norm.sf(abs(z)).

You're asking:

| "What's the chance someone ends up this far from the center on either side?
| To get the total weirdness on both ends of the bridge.

How norm.cdf(z) and norm.sf(z) works internally.

* Exercise Problem

Let's say we have a std normal distribution (mean = 0, standard dev = 1).

Calculate the following for z = 1.5

(1) PDF
(2) CDF
(3) SF (survival function).
(4) Two-Tailed P-value.

(1) PDF:

The height of the normal curve at z = 1.5.

Formula:

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

$$\phi(1.5) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(1.5)^2}{2}} = 0.1295$$

python code

```
from scipy.stats import norm.
pdf = norm.pdf(1.5).
```

(2) CDF

Probability that $z \leq 1.5$

$$\phi(1.5) = \int_{-\infty}^{1.5} \phi(t)\, dt$$

calculation:

From z-table or python

$$\phi(1.5) \approx 0.9332$$

python code:

```
cdf = norm.cdf(1.5) # output: 0.9332
```

(3) Survival function (SF).

probability that $Z > 1.5$

$$SF(1.5) = 1 - \Phi(1.5)$$

$$SF(1.5) = 1 - 0.9332 = 0.0668$$

python code

```
sf = norm.sf(1.5)
```

(4) Two-tailed p-value

Definition

probability of observing $|Z| \geq 1.5$ (both tails combined).

$$p\text{-value} = 2 \times SF(|1.5|)$$

$$p\text{-value} = 2 \times 0.0668 = 0.1336$$

python code

```
p-value = 2 * norm.sf(abs(1.5))
```

(3) Z-test for Proportions.

(1) Theoritical background.

(A) One-sample Z-test for Proportions.

used when you want to test whether a sample proportion is significantly different from a known or claimed population proportion.

Formula:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

where:

- $\hat{p}$: sample proportion
- p_0 : hypothesized population proportion (null hypothesis).
- n : sample size.

(B) Two-sample Z-Test for proportion

used to compare proportions from two independent groups (e.g. conversion rate of A vs B).

where:

- $p_1 = \frac{x_1}{n_1}$, $\hat{p}_2 = \frac{x_2}{n_2}$
- $p = \frac{x_1 + x_2}{n_1 + n_2}$; pooled proportions.

2. Real-world Examples

(A) One-sample Example:

A website claims 60% of visitors sign up for a newsletter. In your sample of 1000 visitors, only 50 signed up. Test the claim at $\alpha = 0.05$.

Given,

- $p_0 = 0.6$
- $\hat{p} = 0.5$
- $n = 1000$

(B) Two-sample Example.

Group A: 80 out of 200 people liked the new product.

Group B: 60 out of 150 liked it.

Is the difference statistically significant?

(3) Python code (A) one same proportion test

```
from scipy.stats import norm
import numpy as np

p_hat = 0.5  # observed proportion
p_0 = 0.6  # hypothesized proportion,
n = 1000
```

#2 - calculation

2-score = (p_hat - p_0) / np.sqrt(p_0 * (1 - p_0 / n))

p-value = 2 * norm.sf(abs(2-score))

print(f"One-sample 2-score: {2-score: .3f}")
print(f"One-sample p-value: {p-value: .3f}").

(B) Two-sample Proportion test.

x1, n1 = 80, 200
x2, n2 = 60, 150
p1 = x1/n1
p2 = x2/n2
p = (x1 + x2) / (n1 + n2) # pooled proportion.

2 calculation

2-score = (p1 - p2) / np.sqrt(p * (1-p) * (1/n1 + 1/n2)).

p-value = 2 * norm.sf(abs(2-score))

print(f"Two-sample 2-score: {2-score: .3f}").
print(f"Two-sample p-value: {p-value: .3f}")

output

- One-Sample Z-score: -2.041
- One-sample P-value: 0.041

Interpretation

Hypothesis Setup:

- Null Hypothesis (H_0): The true proportion is $p_0 = 0.6$.
- Alternative Hypothesis (H_1): The true proportion is not 0.6 ($\neq$) → this is a two-tailed test.

Your Results:

- Sample proportion ($\hat{p}$) = 0.50
- Z-score = -2.041.
  - This means your sample proportion (0.50) is 2.04 standard deviation below the hypothesized mean (0.60).

- P-value = 0.041
  - This means there's a 4.1% chance you'd get a result this extreme if the true proportion really were 0.6.

Interpretation (assuming $\alpha = 0.05$):

- Since p-value = 0.041 < 0.05, the result is statistically significant.
- Conclusion: Reject the null hypothesis.

- Interpretation in simple terms:

Based on the sample data, there's strong enough evidence to suggest that the true proportion is different from 0.6.

Output

Two-sample Z-score: 0.000.
Two-sample P-value: 1.000.

* Hypothesis setup:

- Null Hypothesis (H_0):
  There is no difference in population proportions $\rightarrow p_1 = p_2$

- Alternative Hypothesis (H_1)
  There is a difference $\rightarrow p_1 \neq p_2$ (two-tailed test).

* Sample Details

- Group 1: 80 out of 200 $\rightarrow p_1 = 0.4$.
- Group 2: 60 out of 150 $\rightarrow p_2 = 0.4$

- Pooled proportion $p = \frac{80+60}{200+150} = \frac{140}{350} = 0.4$

* Results:

- 2-score: 0.000 →

Your two sample proportions are exactly equal, so the observed diff is 0 std dev away from what we expected under Ho.

- P-value = 1.000 →

This means the data is perfectly in line with the null hypothesis

There is a 100% probability of observing this (or something more extreme) if Ho is true.

* Interpretation (assuming $\alpha = 0.05$):

- Since p-value = 1.000 > 0.05, this result is not statistically significant.
- Conclusion: Fail to reject the null hypothesis
- Plain English: There is no evidence that the ~~population~~ proportions are different between the two groups.

Real-world analogies for both.

Proportion

One-sample z-test.

Imagine you claim that 60% of voters support a candidate. You survey 100 people and find only 50% support. The math shows this difference is unlikely to be due to random chance, so you're likely wrong about the 60%.

Two-sample z-test.

Imagine you're comparing two sales teams (Team A and Team B) who each close 40% of their leads. Since their performance is identical, the stats confirm that there's no reason to think one team is doing better than the others.

Motive of Hypothesis

Your Analysis

→ Our motive is anyhow to reject the null hypothesis and give preference to the alternative hypothesis.

② Not exactly.

What hypothesis testing is really about:

The goal is not to force rejection of the null hypothesis. Instead, it's to assess whether the evidence from the data is strong enough to justify rejecting it.

Think of this way.

You are judge of scientific research and the null hypothesis is the default assumption, like "the accused is innocent". You only reject it (declare guilt) if the evidence (data) is statistically strong - i.e, the p-value is small.

So, we follow this rule:

- If p-value $< \alpha$ (e.g. 0.05) →
  Evidence is strong enough → ✗ Reject H_0 → ✓ support H_1.

- If p-value $\geq \alpha$ →
  Not enough evidence → ☑ fail to reject H_0 → ☒ cannot support H_1 (yet).

Important:

- You're not trying to prove the alternative hypothesis.
- You're testing if H_0 is unlikely, based on data.
- You should stay neutral and let the data speak - like a judge in court.

Real-world analogy:

Imagine you're a weather forecaster.

- H_0: No storm is coming
- H_1: A storm is coming
- You get some radar data (your sample).
- If the data is statistically extreme (low p-value), you issue a storm warning.
- If not, you say: "Nothing unusual, probably no storm."

p-value syntax for Manual and Built-in function

Z-test

(1) One-sample Z-test

(a) Manually

Step 1 #

```
z-stat = (sample-mean - population-mean)
         / (population-std / np.sqrt(
           sample-size))

# two-tails
p-value = 2 * norm.sf(abs(z-stat))

# left-tail
p-value = norm.cdf(z-stat).

# Right-tail.
p-value = norm.sf(z-stat)
```

(b) Built-in function.

```
from statsmodels.stats.weightstats import
    ztest

# z-test using the built-in function (two-tailed
    p-value).
z-stat, p-two-tailed = ztest(random-sample
    , value = population-mean)

# left-tail p-value (one-tailed).
p_left = p-two-tailed / 2 if z-stat < 0
         else 1 - (p-two-tailed/2).
```

p_right = p_two_tailed/2 if z_stat > 0 else
1 - (p_two_tailed/2).

OR,

two_tails.
z_stat, p_two_tailed = ztest(random_sample,
value = population_mean, alternative =
'two_sided')
left_tails (H1: sample mean < pop mean)
z_stat, p_left = ztest(random_sample,
value = population_mean, alternative
= 'smaller')
right_tails (H2: sample mean > pop mean)
z_stat, p_right = ztest(random_sample,
value = population_mean, alternative.
= 'larger')

(2) Two-Sample z-test.

(a) Manual

Sample as one sample (for p_value).

(b) Built in function.

from statsmodels.stats.weightstats import
ztest.

Two_Tailed test (default).
z_stat, p_two_tailed = ztest(sample1,
sample2, alternative = 'two_sided')

Left-tailed test (mean of sample1 < mean of sample2).

2-stat, p-left = 2test (sample1, sample2, alternative = 'smaller').

Right-tailed test (mean of sample1 > mean of sample2).

2-stat, p-right = 2test (sample1, sample2, alternative = 'larger').

(3) 2-Test for Proportions.

(1) One-sample 2-test for proportion.

(a) Manually.

same / same.

(b) Built in functions

Two-Tailed Test:

2-stat, p-two-tailed = proportion-2test (count, nobs, value = p-0, alternative = 'two-sided')

Right Tailed Test: $H_0: p \leq p_0$ vs $H_1: p > p_0$.

2-stat, p-right = proportion-2test (count, nobs, value = p-0, alternative = 'larger')

Left Tailed Test: $H_0: p \geq p_0$ vs $H_1: p < p_0$.

2-stat, p-left = proportion-2test (count, nobs, value = p-0, alternative = 'smaller').

p-value syntax

t-test

1. One Sample t-test.

(a) Manually

Two-Tailed Test.

p-value = 2 * stats.t.sf(np.abs(t-stat), df).

Left-Tailed test.

$H_0: \mu \geq \mu_0$ vs $H_1: \mu < \mu_0$.

p-left = stats.t.cdf(t-stat, df).

Right-Tailed Test:

$H_0: \mu \leq \mu_0$ vs $H_1: \mu > \mu_0$.

p-right = stats.t.sf(t-stat, df).

(b) Built-in function

Two-Tailed,

t-stat, p-value = stats.ttest_1samp(sample, popmean = mu_0).

For One-Tailed.

→ The ttest_1samp function does not directly support one-tailed tests with an alternative parameter.

Left-tailed

if t-stat < 0:

p-value_left = p-value / 2.

Right-tailed.

if t-stat > 0

p-value_right = p-value / 2.

(2) Independent two sample test.

(a) Manually,

The formulas for calculating the two tailed, left-tailed and right tailed p-values remain the same for all types of t-test.

(b) Built in functions.

Two-tailed test.

t-stat, p-value-two-tailed = stats.ttest_ind(group1, group2, equal_var=True, alternative='two-sided').

Left-tailed.

t-stat, p-value-left = stats.ttest_ind(group1, group2, equal_var=True, alternative='less').

Right-tailed.

t-stat, p-value-right = stats.ttest_ind(group1, group2, equal_var=True, alternative='greater').

(4) Paired test.

(b) Built in functions

Two-tailed test

```
t-stat, p-value-two-tailed = stats.ttest
        -rel(after, before, alternative=
        'two-sided')
```

left tailed test.

```
t-stat, p-value-left = stats.ttest-rel
        (after, before, alternative =
         'less').
```

Right tailed test.

```
t-stat, p-value-right = stats.ttest-rel
        (after, before, alternative =
         'greater').
```